List of Suggested Projects (MATP32)

Projects related to the Hardy–Littlewood maximal operator

1. Stein's $L \log L$ theorem for the classical maximal operator

Let M denote the classical centred Hardy–Littlewood maximal operator formed with respect to balls in $\mathbb{R}^d$.

As we have seen in the lectures, for every non-zero function $f \in L^1(\mathbb{R}^d)$ one has that $M(f) \notin L^1(\mathbb{R}^d)$. In [32], E. M. Stein showed that, for any given finite ball $B \subset \mathbb{R}^d$ and a measurable function f supported in B,

$$M(f) \in L^1(B) \quad \Longleftrightarrow \quad \int_B |f(x)| \log(e + |f(x)|)\, dx < \infty. \tag{1}$$

The goal of this project is to present E. M. Stein's paper [32], which contains (1) and an application of (1) concerning local integrability of Riesz transforms on $\mathbb{R}^d$.

Suggested literature.

- Stein, Elias M. "Note on the class $L \log L$." *Studia Math.* 32, no. 3 (1969), 305–310.

2. Fefferman–Stein inequalities

Let M^0 denote the classical centred Hardy–Littlewood maximal operator formed with respect to cubes in $\mathbb{R}^d$.

As we have seen in the lectures, M^0 is of weak type $(1,1)$ and bounded on $L^p(\mathbb{R}^d)$ for $p > 1$. The goal of this project is to present a paper of C. Fefferman and E. M. Stein in which certain vector-valued extensions of the aforementioned results are obtained.

Suggested literature.

- Fefferman, C.; Stein, E. M. "Some maximal inequalities." *Amer. J. Math.* 93 (1971), 107–115.

3. Weighted inequalities for maximal function

Let w be a locally integrable nonnegative function on $\mathbb{R}^d$ (such a function is called a weight), let $p \in (1, \infty)$ and let

$$L^p_w(\mathbb{R}^d) = \left\{ f : \mathbb{R}^d \to \mathbb{R} \text{ measurable} : \int_{\mathbb{R}^d} w(x) |f(x)|^p dx < \infty \right\}.$$

Recall that the (unweighted) Hardy-Littlewood maximal function, given by

$$Mf(x) = \sup_{Q \text{ cube}: x \in Q} \frac{1}{|Q|} \int_Q |f(y)| dy,$$

is a bounded sublinear operator on $L^p(\mathbb{R}^d)$. The question we want to pursue in this project is for which weights w, M is also a bounded sublinear operator on $L^p_w(\mathbb{R}^d)$. Muckenhoupt proved in 1972 [23] that this is the class of weights characterised by the so-called Muckenhoupt A_p characteristic,

$$[w]_{A_p} := \sup_{Q \text{ cube}} \frac{1}{|Q|}\int_Q w(x)dx \frac{1}{|Q|}\int_Q w^{-1/(p-1)}(x)dx < \infty.$$

Lerner gave a much shorter proof in [21], Theorem C. Here, he even proved a precise estimate on how $\|M\|_{L^p_w \to L^p_w}$ depends on the characteristic $[w]_{A_p}$ (this was formerly proven by S. Buckley). The goal of the project is to present a proof of Theorem C in [21].

Suggested literature.

- A. K. Lerner, "An elementary approach to several results on the Hardy-Littlewood maximal operator", Proc. Amer. Math. Soc. **136** (2008), no. 8, 2829–2833.

Projects related to certain classical maximal principles

4. Calderón 's maximal principle

Recall that if $f \in L^1(\mathbb{T})$, then its Fourier coefficient $\widehat{f}(n)$ at $n \in \mathbb{Z}$ is given by

$$\widehat{f}(n) := \int_{\mathbb{T}} f(x)e^{-i2\pi nx}\,dx.$$

For $N \in \mathbb{N}$, the N-th partial sum $S_N(f)$ of the Fourier series of f is the trigonometric polynomial

$$S_N(f)(x) := \sum_{n=-N}^{N} \widehat{f}(n)e^{i2\pi nx}, \qquad x \in \mathbb{T}.$$

In 1966, in [5], L. Carleson confirmed a conjecture of N. Lusin showing that if $f \in L^2(\mathbb{T})$, then $S_N(f)(x) \to f(x)$ as $N \to \infty$ for a.e. $x \in \mathbb{T}$.

Before L. Carleson's resolution of N. Lusin's conjecture, A. P. Calderón had shown that for $f \in L^2(\mathbb{T})$, $S_N(f)$ converges to f a.e. if, and only if, the maximal operator

$$C(f)(x) := \sup_{N\in\mathbb{N}} |S_N(f)(x)|$$

is of weak type $(2,2)$. The goal is to present this result of A. P. Calderón.

Suggested literature.

- §2.1 in de Guzmán, Miguel. *Real Variable Methods in Fourier Analysis*. North-Holland Mathematics Studies, 46. North-Holland, Amsterdam–New York, 1981.

5. Stein's maximal principle

The goal is to present an important maximal principle of E. M. Stein [33], which in the case of $\mathbb{T}$ can be stated as follows. Let $p \in [1,2]$ be given and let $\{T_k\}_{k\in\mathbb{N}}$ be a sequence of translation-invariant operators that are bounded on $L^p(\mathbb{T})$. Consider the maximal operator

$$T(f)(x) := \sup_{k\in\mathbb{N}} |T_k(f)(x)|.$$

Then the following are equivalent:

1. for each $f \in L^p(\mathbb{T})$, $T(f)(x) < \infty$ for a.e. $x \in \mathbb{T}$;

2. T is of weak type (p, p).

Suggested literature.

- §2.3 in de Guzmán, Miguel. *Real Variable Methods in Fourier Analysis.* North-Holland Mathematics Studies, 46. North-Holland, Amsterdam–New York, 1981.
- §7–9 in Stein, E. M. "On limits of sequences of operators." *Ann. of Math.* (2) 74 (1961), 140–170.

Projects related to classical Littlewood–Paley theory and its applications

6. The Marcinkiewicz multiplier theorem on $\mathbb{R}$

Let m be a function in $L^\infty(\mathbb{R})$. We say that m is a Marcinkiewicz multiplier on $\mathbb{R}$ if, moreover, m is C^1 in all intervals of the form $(2^j, 2^{j+1})$ or $(-2^{j+1}, -2^j)$, $j \in \mathbb{Z}$, and

$$\sup_{j\in\mathbb{Z}} \left[\int_{-2^{j+1}}^{-2^j} |m'(\xi)|\, d\xi + \int_{2^j}^{2^{j+1}} |m'(\xi)|\, d\xi \right] < \infty.$$

Consider the corresponding operator $T_m(f) := \check{m} * f$, $f \in \mathcal{S}(\mathbb{R})$. The goal is to present a proof of the fact that T_m can be extended as an L^p-bounded operator for all $1 < p < \infty$.

Suggested literature.

- §6.1.3, §6.2.1 in Grafakos, Loukas. *Classical Fourier Analysis.* Third edition. Graduate Texts in Mathematics, 249. Springer, New York, 2014.

7. Rubio de Francia's theorem

If I is a collection of intervals in $\mathbb{R}$, consider the corresponding Littlewood–Paley operator

$$S_I(f) := \left(\sum_{I\in\mathcal{I}} |\check{\chi}_I * f|^2 \right)^{1/2}.$$

The classical Littlewood–Paley theorem asserts that if

$$\mathcal{I} = \{\pm[2^j, 2^{j+1}) : j \in \mathbb{Z}\},$$

then S_I is bounded on $L^p(\mathbb{R})$ for all $1 < p < \infty$.

In 1984, J. L. Rubio de Francia showed in [29] that for any collection I of intervals in $\mathbb{R}$, S_I is bounded on $L^p(\mathbb{R})$ for all $p \in [2, \infty)$. The goal of this project is to present the proof of the aforementioned theorem.

Suggested literature.

- Rubio de Francia, José L. "A Littlewood–Paley inequality for arbitrary intervals." *Rev. Mat. Iberoamericana* 1 (1985), no. 2, 1–14.

Projects related to the space BMO

8. Sharp John-Nirenberg constants on $\mathbb{R}$

The John-Nirenberg Theorem states that there exists $C_1, C_2 > 0$ such that for any $b \in \mathrm{BMO}(\mathbb{R})$, any interval I in $\mathbb{R}$ and any $\lambda > 0$,

$$|\{x \in I : |b(x) - \langle b \rangle_I| > \lambda\}| \leq C_1 |I| e^{-C_2 \lambda / \|b\|_{\mathrm{BMO}}}$$

In [20], it was shown that the largest possible choice of C_2 is $C_2 = 2/e$. A. Lerner showed in 2013 that for this choice of C_2, the smallest choice of C_1 is $C_1 = \frac{1}{2} e^{4/e}$. The goal of the project is to present Lerner's paper.

Suggested literature.

- A. K. Lerner, "The John-Nirenberg inequality with sharp constants", C. R. Math. Acad. Sci. Paris **351** (2013), no. 11-12, 463–466.

9. Carleson's counterexample

Let $\mathcal{I}$ denote the set of all dyadic intervals in $[0, 1)$,

$$\mathcal{I} := \{[2^{-k}m, 2^{-k}(m+1)) : k, m \in \mathbb{N}_0\}.$$

For $I \in \mathcal{I}$, let h_I denote the corresponding Haar function adapted to I:

$$h_I := |I|^{-1/2}(\chi_{I^-} - \chi_{I^+}),$$

where I^- and I^+ denote the left and right halves of I, respectively. If $I, J \in \mathcal{I}$, write

$$h_{I \times J}(x, y) := h_I(x) h_J(y).$$

The two-parameter dyadic BMO space $\mathrm{BMO}_{\mathrm{prod}}([0,1)^2)$ is defined as the class of all functions $b \in L^1([0,1)^2)$ for which there exists a constant $C_b > 0$ such that

$$\sum_{I \times J \subset \Omega} |\langle b, h_{I \times J} \rangle|^2 \leq C_b |\Omega| \qquad \text{for all open } \Omega \subseteq [0,1)^2. \tag{5}$$

The two-parameter dyadic rectangular BMO space $\mathrm{BMO}_{\mathrm{rect}}([0,1)^2)$ is defined as the class of all functions $b \in L^1([0,1)^2)$ for which there exists a constant $C_b > 0$ such that

$$\sum_{I \times J \subset R} |\langle b, h_{I \times J} \rangle|^2 \leq C_b |R| \qquad \text{for all } R \in \mathcal{I} \times \mathcal{I}. \tag{6}$$

The goal is to present a construction due to L. Carleson [4], following the presentation in §7 in T. Tao's notes, which shows that

$$\mathrm{BMO}_{\mathrm{prod}}([0,1)^2) \not\subset \mathrm{BMO}_{\mathrm{rect}}([0,1)^2).$$

Suggested literature.

- Tao, Terence. "Dyadic product H^1, BMO, and Carleson's counterexample." Available at http://www.math.ucla.edu/~tao/preprints/harmonic.html.

Projects in dyadic harmonic analysis

10–11. The Chang–Wilson–Wolff inequality

Let $\mathcal{I}$ denote the set of all dyadic intervals in $[0,1)$,

$$\mathcal{I} := \{[2^{-k}m, 2^{-k}(m+1)) : k, m \in \mathbb{N}_0\}.$$

For $f \in L^1([0,1))$, consider the corresponding sequence of dyadic martingale differences $\{D_k[f]\}_{k\in\mathbb{N}_0}$ by

$$D_k[f] := \begin{cases} E_k[f] - E_{k-1}[f], & k \geq 1, \\ E_0[f], & k = 0, \end{cases}$$

where

$$E_k[f] := \sum_{I\in\mathcal{I}:\, |I|=2^{-k}} \left(\frac{1}{|I|}\int_I f(y)\,dy\right)\chi_I.$$

Consider the dyadic square function

$$S[f] := \left(\sum_{k\geq 0} |D_k[f]|^2\right)^{1/2}.$$

A celebrated result of S.-Y. A. Chang, J. M. Wilson, and T. H. Wolff (see [6]) asserts that

$$S[f] \in L^\infty([0,1)) \quad \Longrightarrow \quad \int_{[0,1)} e^{c|f(x)|^2}\,dx < \infty \quad \text{for some } c > 0. \tag{2}$$

A different proof of the implication (2) was given by L. Slavin and A. Volberg [30] using Bellman function techniques.

There are two projects related to the theorem of Chang, Wilson, and Wolff:

- Present the proof of (2) appearing in [6].
- Present the proof of (2) appearing in [30].

Suggested literature.

- §3 in Chang, S.-Y. A.; Wilson, J. M.; Wolff, T. H. "Some weighted norm inequalities concerning the Schrödinger operators." *Comment. Math. Helv.* 60 (1985), no. 2, 217–246.
- Slavin, Leonid; Volberg, Alexander. "The S-function and the exponential integral." In *Topics in Harmonic Analysis and Ergodic Theory*, 215–228, Contemp. Math., 444, AMS, 2007.

12. Duality between dyadic H^1 and dyadic BMO via Bellman function techniques

The goal is to present a proof due to L. Slavin and A. Volberg [31] of the duality between dyadic H^1 and dyadic BMO based on Bellman function techniques.

Suggested literature.

- Slavin, Leonid; Volberg, Alexander. "Bellman function and the H^1–BMO duality." In *Harmonic Analysis, Partial Differential Equations, and Related Topics*, 113–126, Contemp. Math., 428, AMS, 2007.

13. John–Nirenberg inequality via Bellman function techniques

The goal is to present a proof of T. Tao of the John–Nirenberg inequality for the dyadic BMO space via Bellman function techniques.

Suggested literature.

- Tao, Terence. "Bellman functions and the John–Nirenberg inequality." Available at http://www.math.ucla.edu/~tao/preprints/harmonic.html.

14. Petermichl's dyadic shift representation for the Hilbert transform

S. Petermichl used in her PhD thesis in 2000 the so-called dyadic shift Ш to represent the Hilbert transform as an average of much simpler operators [26]. Here, Ш is defined on the Haar basis $\{h_I\}_{I\in\mathcal{D}}$ of $L^2(\mathbb{R})$ by

$$Шh_I = h_{I^+} - h_{I^-} \quad (I \in \mathcal{D}),$$

where I^-, I^+ denote the left and right half of I, respectively. In 2008, T. Hytönen gave a simplified proof of this representation with the precise constant [16]. The goal of the project to present Hytönen's proof of Petermichl's representation of the Hilbert transform.

Suggested literature.

- T. P. Hytönen, On Petermichl's dyadic shift and the Hilbert transform, C. R. Math. Acad. Sci. Paris **346** (2008), no. 21-22, 1133–1136.
- §5.1 in: T. P. Hytönen et al., *Analysis in Banach spaces. Vol. I. Martingales and Littlewood-Paley theory*, Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge. A Series of Modern Surveys in Mathematics, 63, Springer, Cham, 2016.

Projects related to the Kakeya problem

15. The Kakeya problem on $d = 2$

A Kakeya set is a compact set $K \subset \mathbb{R}^d$ that contains a unit line segment in every direction. The Kakeya conjecture asks whether any Kakeya set in $\mathbb{R}^d$ has Hausdorff dimension d. In [7], R. O. Davies solved the conjecture in the two-dimensional case. In three dimensions, it was proved by Hong Wang and Joshua Zahl in 2026 [37]. OpenAI claimed in their release on October 7, 2026 that ChatGPT had solved the case of dimension 4.

The goal of this project is to present a proof of the fact that the so-called Kakeya maximal function conjecture implies the Kakeya conjecture and then present J. Bourgain's proof of the Kakeya maximal function conjecture on $\mathbb{R}^2$ [1], following the presentation in §10 of T. H. Wolff's book.

Suggested literature.

- Wolff, Thomas H. *Lectures on Harmonic Analysis.* University Lecture Series, 29. American Mathematical Society, 2003.

16. Dvir's resolution of the finite-field Kakeya conjecture

Motivated by the unsolved Kakeya conjecture for $d \geq 3$, T. H. Wolff proposed a model version of the problem on finite fields. More specifically, given a finite field $\mathbb{F}$, a set $K \subseteq \mathbb{F}^d$ is said to be a Kakeya set if and only if K contains at least one affine line parallel to each direction in $\mathbb{F}^d \setminus \{0\}$.

The finite-field Kakeya conjecture states that there exists a constant $C_d > 0$ such that for every Kakeya set $K \subseteq \mathbb{F}^d$ one has

$$\#(K) \geq C_d \, [\#(\mathbb{F})]^d.$$

The goal is to present Z. Dvir's resolution of the conjecture [9].

Suggested literature.

- Dvir, Zeev. "On the size of Kakeya sets in finite fields." *J. Amer. Math. Soc.* 22 (2009), no. 4, 1093–1097.
- Wolff, Thomas. "Recent work connected with the Kakeya problem." In *Prospects in Mathematics*, 129–162, AMS, 1999.

Projects related to analytic Hardy spaces

17.–19. Marcinkiewicz-type decomposition for analytic Hardy spaces

Let $r < p < q$ and let $g \in L^p$. Recall that in the proof of the Marcinkiewicz interpolation theorem, for $\alpha > 0$, one decomposes g as

$$g = g\chi_{\{|g|>\alpha\}} + g\chi_{\{|g|\leq\alpha\}} \in L^r + L^q.$$

In certain problems, it is of interest to obtain analogous decompositions for functions in analytic Hardy spaces, where, for $p \geq 1$, the analytic Hardy space $H^p_A(\mathbb{T})$ can be defined as the set of functions $f \in L^p(\mathbb{T})$ such that $\widehat{f}(n) = 0$ for $n < 0$. Unfortunately, if $f \in H^p_A(\mathbb{T})$ and $\alpha > 0$, then $f\chi_{\{|f|>\alpha\}} \notin H^r_A(\mathbb{T})$ and $f\chi_{\{|f|\leq\alpha\}} \notin H^q_A(\mathbb{T})$ in general. However, in [17, **?**], P. W. Jones obtained an appropriate substitute of Marcinkiewicz decomposition for analytic Hardy spaces.

Some years later, J. Bourgain, S. V. Kislyakov and Q. Xu, G. Pisier, and P. F. X. Müller obtained different proofs of P. W. Jones's Marcinkiewicz-type decomposition. There are several possible projects related to the discussion above:

- Briefly introduce analytic Hardy spaces on $\mathbb{T}$ and then present the Marcinkiewicz-type decomposition due to S. V. Kislyakov and Q. Xu, following, e.g., §7.4 in Pavlović.
- Briefly introduce analytic Hardy spaces on $\mathbb{T}$ and then present the Marcinkiewicz-type decomposition due to J. Bourgain and in particular, §1 in [2].
- Briefly introduce analytic Hardy spaces on $\mathbb{R}$ and then present P. W. Jones's approach for decomposing functions in analytic Hardy spaces on the real line, in particular Theorem 2 in [17]. Alternatively, one can follow the presentation in the cited alternative reference.

Suggested literature.

- Bourgain, J. "New Banach space properties of the disc algebra and H^∞." *Acta Math.* 152 (1984), no. 1–2, 1–48.
- Jones, Peter W. "L^∞ estimates for the $\bar{\partial}$ problem in a half-plane." *Acta Math.* 150 (1983), no. 1–2, 137–152.
- Jones, Peter W. "On interpolation between H^1 and H^∞." In *Interpolation Spaces and Allied Topics in Analysis* (Lund, 1983), 143–151, Lecture Notes in Math., 1070, Springer, 1984.
- Kislyakov, S. K.; Xu, Q. "Real interpolation and singular integrals." *Algebra i Analiz* 8 (1996), no. 4, 75–109; translation in *St. Petersburg Math. J.* 8 (1997), no. 4, 593–615.
- Müller, Paul F. X. "Holomorphic martingales and interpolation between Hardy spaces." *J. Anal. Math.* 61 (1993), 327–337.

- Pavlović, Miroslav. *Introduction to Function Spaces on the Disk*. Posebna Izdanja [Special Editions], 20. Matematički Institut SANU, Belgrade, 2004.

20. Sharp Hardy-Littlewood theorem and isoperimetric inequality

The isoperimetric inequality states that among all domains in the plane which can be enclosed by a simple rectifiable curve of fixed length, a disc has the largest area. This theorem has many proofs, but the first proof by means of complex analysis was obtained in 1921 by T. Carleman in 1921 [3]. In 2003, D. Vukotic [36] generalized Carleman's method with fairly elementary methods to prove a form of a classical theorem of Hardy and Littlewood [15] relating Hardy- and Bergman spaces with precise constants. As a corollary, one obtains the isoperimetric inequality.

The goal of the project is to first briefly introduce analytic Hardy- and Bergman spaces on the unit disk, and to then present Vukotic' result.

Suggested literature.

- D. Vukotić, "The isoperimetric inequality and a theorem of Hardy and Littlewood", Amer. Math. Monthly **110** (2003), no. 6, 532–536.

Miscellaneous projects

21. Yano-type extrapolation

Let $(X, \mathcal{A}, \mu)$ and $(Y, \mathcal{B}, \nu)$ be two finite measure spaces and let T be an operator that maps $\mathcal{A}$-measurable functions to $\mathcal{B}$-measurable functions. If there exist absolute constants $C, r > 0$ and an exponent $p_0 > 1$ such that

$$\|T\|_{L^p(X)\to L^p(Y)} \leq C(p-1)^{-r} \qquad \text{for all } p \in (1, p_0],$$

then there exists a constant C', depending on C, r, p_0, such that

$$\|T(f)\|_{L^1(Y)} \leq C' \left[1 + \int_X |f(x)| \log^r(e + |f(x)|)\, d\mu(x)\right].$$

Suggested literature.

- Yano, Shigeki. "Notes on Fourier analysis. XXIX. An extrapolation theorem." *J. Math. Soc. Japan* 3 (1951), 296–305.
- §4 in Chapter XII of Zygmund, Antoni. *Trigonometric Series*. Vols. I–II. Cambridge University Press, 2002.

22.–23. Best constants in M. Riesz's theorem on the Hilbert transform

Let H denote the Hilbert transform acting on functions defined over the real line. In 1972, S. K. Pichorides showed that

$$\|H\|_{L^p\to L^p} = \begin{cases} \tan\left(\dfrac{\pi}{2p}\right), & p \in (1, 2], \\ \cot\left(\dfrac{\pi}{2p}\right), & p \in [2, \infty). \end{cases} \tag{3}$$

A different proof of (3) was given in the references listed below. There are two projects related to Pichorides's theorem:

- Present the proof of (3) appearing in Grafakos.
- Present an alternative proof of (3) appearing in Grafakos–Savage.

Suggested literature.

- Grafakos, Loukas. "Best bounds for the Hilbert transform on $L^p(\mathbb{R}^1)$." *Math. Res. Lett.* 4 (1997), no. 4, 469–471.
- Grafakos, Loukas; Savage, Thomas. "Best bounds for the Hilbert transform on $L^p(\mathbb{R}^1)$: a corrigendum." *Math. Res. Lett.* 22 (2015), no. 5, 1333–1335.
- Pichorides, S. K. "On the best values of the constants in the theorems of M. Riesz, Zygmund and Kolmogorov." *Studia Math.* 44 (1972), 165–179.

24. Lacunary Fourier series

Recall that if $f \in L^1(\mathbb{T})$, then its Fourier coefficient $\widehat{f}(n)$ at $n \in \mathbb{Z}$ is given by

$$\widehat{f}(n) := \int_{\mathbb{T}} f(x) e^{-i2\pi nx}\, dx.$$

A classical result due to A. Zygmund [43] asserts that if f is a trigonometric polynomial with $\operatorname{supp}(\widehat{f}) \subset \{3^n\}_{n\in\mathbb{N}}$, so that

$$f(x) = \sum_{n=1}^{N} a_n e^{i2\pi 3^n x},$$

then

$$\|f\|_{L^p(\mathbb{T})} \leq C_0 p^{1/2} \|f\|_{L^2(\mathbb{T})} \qquad \text{for all } p \in [2,\infty), \tag{4}$$

where $C_0 > 0$ is an absolute constant. Moreover, the exponent $1/2$ in $p^{1/2}$ is best possible. The goal of this project is to present the proof of (4), following Chapter V, §7–8 of Zygmund's *Trigonometric Series*. This project involves Riesz products, Khintchine's inequality, and Young's convolution inequality.

Suggested literature.

- Zygmund, Antoni. *Trigonometric Series*. Vols. I–II. Cambridge University Press, 2002.

25. Zygmund's theorem on two-dimensional restriction

This project is concerned with A. Zygmund's proof [41] of the so-called restriction conjecture in the two-dimensional case. More specifically, if $f \in L^1(\mathbb{T}^2)$ then its Fourier coefficient at $(m,n) \in \mathbb{Z}^2$ is

$$\widehat{f}(m,n) := \int_{\mathbb{T}^2} f(x,y) e^{-i2\pi(mx+ny)}\, dx\, dy.$$

The goal is to present Zygmund's proof of the fact that if $p \in [1, 4/3]$ and $q = p'/2$, then

$$\left(\sum_{(m,n)\in\mathbb{Z}^2 :\, m^2+n^2=R^2} |\widehat{f}(m,n)|^q \right)^{1/q} \leq A_p \|f\|_{L^p(\mathbb{T}^2)}, \tag{7}$$

for all $R > 0$, where the constant may be taken to be $A_p = 5^{1/p}$.

Suggested literature.

- §1–3 in Zygmund, A. "On Fourier coefficients and transforms of functions of two variables." *Studia Math.* 50 (1974), 189–201.

26. The multiplier problem for the ball

Let B denote the unit ball in $\mathbb{R}^d$, $d \geq 2$, and let

$$T(f) := \check{\chi}_B * f, \qquad f \in \mathcal{S}(\mathbb{R}^d).$$

The goal is to present a celebrated result due to C. Fefferman [10], which asserts that T is unbounded on $L^p(\mathbb{R}^d)$ for any $p \in (1,2) \cup (2,\infty)$.

Suggested literature.

- Fefferman, Charles. "The multiplier problem for the ball." *Ann. of Math.* (2) 94 (1971), 330–336.

Marking criteria for the projects in MATP32

The length of the project should be 6-12 pages in LaTeX. **Total score: 40 points**

Written part: 20 p

- 5p Introduction: Connection with the contents of the course (if applicable), context, motivation
- 5p Structure and coherence of the whole text, avoidance of notational or other inconsistencies (including typos and minor errors).
- 10p A clear and rigorous presentation of the proof of the main result.

Feedback: 5 hp

- 5p Meaningful and constructive feedback to your writing partner (about half a page)

Oral Presentation: 15 hp

- 5p Structure and coherence of the presentation
- 5p Clarity of exposition/discussion
- 5p Contact with audience, answer to questions

References

[1] J. Bourgain, "Besicovitch type maximal operators and applications to Fourier analysis," *Geom. Funct. Anal.* 1 (1991), no. 2, 147–187.

[2] J. Bourgain, "New Banach space properties of the disc algebra and H^∞," *Acta Math.* 152 (1984), no. 1–2, 1–48.

[3] T. Carleman, "Zur Theorie der Minimalflächen", Math. Z. **9** (1921), no. 1-2, 154–160.

[4] L. Carleson, *A counterexample for measures bounded on H^p for the bidisc*, Mittag-Leffler Report No. 7, 1974.

[5] L. Carleson, "On convergence and growth of partial sums of Fourier series," *Acta Math.* 116 (1966), 135–157.

[6] S.-Y. A. Chang, J. M. Wilson and T. H. Wolff, "Some weighted norm inequalities concerning the Schrödinger operators," *Comment. Math. Helv.* 60 (1985), no. 2, 217–246.

[7] R. O. Davies, "Some remarks on the Kakeya problem," *Proc. Cambridge Philos. Soc.* 69 (1971), 417–421.

[8] M. de Guzmán, *Real Variable Methods in Fourier Analysis*, North-Holland Mathematics Studies, 46. North-Holland, Amsterdam–New York, 1981.

[9] Z. Dvir, "On the size of Kakeya sets in finite fields," *J. Amer. Math. Soc.* 22 (2009), no. 4, 1093–1097.

[10] C. Fefferman, "The multiplier problem for the ball," *Ann. of Math.* (2) 94 (1971), 330–336.

[11] C. Fefferman and E. M. Stein, "Some maximal inequalities," *Amer. J. Math.* 93 (1971), 107–115.

[12] L. Grafakos, *Classical Fourier Analysis*, third edition, Graduate Texts in Mathematics, 249, Springer, New York, 2014.

[13] L. Grafakos, "Best bounds for the Hilbert transform on $L^p(\mathbb{R}^1)$," *Math. Res. Lett.* 4 (1997), no. 4, 469–471.

[14] L. Grafakos and T. Savage, "Best bounds for the Hilbert transform on $L^p(\mathbb{R}^1)$: a corrigendum," *Math. Res. Lett.* 22 (2015), no. 5, 1333–1335.

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